Mapping intra-urban social deprivation from Sentinel imagery using only municipal aggregate labels

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Abstract

Most countries publish social statistics as administrative aggregates; almost none publish them at neighborhood level. We measure how much neighborhood detail aggregates and free satellite imagery jointly recover. Trained only on five-level deprivation aggregates of Mexican municipalities, with no spatial annotation, and scored against fully held-out census-tract grades, the map reaches a within-municipality rank correlation of $\rho = 0.20$ on fourteen test cities opened once, with frozen models: 84% of what the same features attain there under full tract supervision (0.23; 90% on the validation cities used for selection). More aggregates keep buying map, state aggregates forfeit part of it, one national figure erases it, and optical content matters where resolution does not. The maps close 31–66% of the aggregate-to-census targeting gap on test, replicate against an independent institution's index, beat the incumbent wealth index where selection occurred and match it on test, and order Bogotá's block strata without retraining. Attention-based multiple-instance learning fails here for a measurable, structural reason.

1 Introduction

Census-quality social data is expensive, and most of the world sees it roughly once a decade. In between, and below the level the tables are published at, policy relies on aggregates: municipal indices and district averages. A rich literature has shown that satellite imagery plus machine learning predicts such indicators across space [1–4], and derived global products already steer programme targeting [3, 5]. Predicting an aggregate somewhere else and disaggregating it in place are different tasks, and the second is the one targeting decisions require: a mayor already knows the municipality is poor but lacks a map of which neighborhoods are poorest.

We ask the disaggregation question quantitatively: how much neighborhood detail can be recovered from a country's published aggregates combined with free imagery? We frame the answer as an exchange rate between the number and granularity of published aggregates and the within-municipality detail they recover, and we measure it in a country that publishes ground truth at both levels. Mexico publishes its Social Deprivation Grade (GRS) both as municipal aggregates and for 61,430 urban census tracts (AGEB) [6]; we train only on the former and score maps against the latter, held out entirely, in cities the model never saw (Fig. 1).

Three design choices constrain the measurement. Supervision is weak: a model sees a bag of image tokens for a municipality and one vector, the population shares at each grade threshold, with no spatial annotation of any kind. Every comparison is scored *within* municipalities, because ordering tracts across cities is mostly solved by knowing which city is poor (pooled correlations run near 0.46 against 0.23 within, at the same operating point). And an oracle trained directly on the held-out tract labels bounds what the frozen representation supports, so weak supervision is always reported as a fraction of a measured upper bound.

2 Data

2.1 Imagery

Sentinel-2 L2A and Sentinel-1 RTC annual median composites for 2020 (10 m grid, windowed COG reads from Microsoft Planetary Computer), for 424 boxes covering the 138 cities of the national set and the 213-municipality expansion; city boxes are conurbations and typically span several municipalities.

2.2 Labels

CONEVAL GRS 2020: municipal-level population shares at each of four grade thresholds supervise training; AGEB-level grades are used only for evaluation and for the oracle. Auxiliary evaluation sources: CONAPO IMU 2020 [7], ESA WorldCover 2020 [14], Meta RWI [3], IBGE AGSN 2019 [15], Bogotá's district stratification [16].

3 Methods

3.1 Bags and instances

A bag is a municipality (771 after exclusions); instances are 16-pixel tokens (160 m) encoded by a frozen DOFA foundation model [13], both sensors fused per token. Municipalities contributing ground to any held-out city are excluded from the expansion; an earlier version leaked nine validation municipalities this way, detectable only at bag level, and every affected number was re-measured.

3.2 Model

A two-layer head on standardized frozen features predicts four cumulative-grade probabilities per token; the bag prediction is the mean of token predictions, fit with binary cross-entropy; each token's hidden representation is concatenated with the mean of its 8 grid neighbours (480 m), computed city-wide at inference.

3.3 Protocol

110/14/14 cities for train/validation/test, stratified by size and deprivation, no city split across sets. The test cities were opened exactly once, after all development ended: every configuration retrains with early stopping anchored to the validation cities as during development (deterministic given the seed), or loads its saved weights, and the test bags are only ever scored under `inference` `mode`. Nothing was tuned, selected or iterated against test results. Sweeps select on grouped training folds under bag error only; validation cities are scored once per reported configuration; comparisons are paired by municipality over three seeds; every interval on a within-municipality mean is a city-clustered bootstrap (2,000 resamples), and curve error bars are s.d. over seeds.

3.4 Oracle

The same head trained on tract labels, with the same context, bounds the representation ($\rho = 0.251$ with context, 0.202 without).

3.5 Zero-shot transfer

Bogotá and Rio get the same 2020 composite protocol; tokens are scored by the three final seeds with Mexican standardization statistics, deliberately unadapted. Rio's evaluation is restricted to tokens with WorldCover built fraction ≥ 0.10.

3.6 Sweep

Fourteen configurations varied one ingredient each (learning rate 5e-5–1e-3, hidden width 64–512, dropout 0–0.5, weight decay, loss, input standardization), then a combination round; standardized inputs with binary cross-entropy won on bag error over five grouped training folds.

3.7 Targeting simulation

For each validation city and budget $q \in \{5, 10, 20, 30\}\%$ of population, tracts are funded in the order given by each allocator (municipal aggregate with ties broken at random over 200 draws, the map's tract-mean score, or tract truth), with the marginal tract funded fractionally, so the truth ordering upper-bounds both by construction. The outcome is the population living in high or very-high grade tracts among those reached, pooled over cities before the gap ratio is formed.

3.8 RWI pairing

Each AGEB takes the RWI value of the nearest grid point to its centroid; the model is averaged token-to-tract; both are scored on the 2,958 shared AGEB in the 22 municipalities with at least 20 scored tracts and more than one grade, and the interval is a city-clustered bootstrap of the per-municipality difference.

3.9 MAUP bins

Tracts are binned by tokens contained (1–3, 4–9, 10–19, 20–49, $\geq$50); within-municipality correlations use municipalities with at least ten tracts in the bin.

3.10 Border test

Adjacent token pairs (160 m apart) are classed by what their midpoint crosses; the excess mean absolute score jump of same-grade municipal-border pairs over same-tract pairs is city-bootstrapped; same-grade AGEB-border pairs inside one municipality are the placebo.

3.11 Software and compute

Python 3.12 with PyTorch, rasterio, geopandas and scikit-learn, pinned by the repository's `uv.lock`; all experiments ran on one Apple-silicon laptop (MPS backend). Seeds 0, 1 and 2 throughout. The seven municipalities excluded from the expansion for insufficient composite depth or persistent acquisition errors (Supplementary Table 3) are the only data exclusions.

3.12 Data and code

The full pipeline, frozen vectors, labels, splits and protocol are released; every figure regenerates from one canonical results file.

4 Results

4.1 Supervision efficiency

Figure 2 shows the measurement on a pool of 771 municipal bags: the 110 training cities plus 213 municipalities added expressly to cover the deprived half of urban Mexico that large-city samples miss (Fig. 1a; municipalities overlapping any held-out city's ground are excluded, Methods). Every row below was scored once on the fourteen test cities with frozen, validation-selected models after all development ended; validation values are shown beside them as the selection benchmark. **Quantity** (Fig. 2a): fifty aggregates yield $\rho \approx 0.15$ within municipalities; the full national supply yields 0.196 on test against a fully supervised upper bound of 0.233 there (per-point values in Supplementary Table 1), an 84% recovery (validation: 0.227 of 0.251, 90%). Pairing model and oracle by municipality, the city-clustered bootstrap puts the test fraction at 0.85 with a wide interval [0.30, 1.86]; fourteen cities bound the precision, and it is the point estimate, consistent across splits, that the title claims. The test curve reproduces the rise through 400 aggregates almost exactly and flattens at the top: the final gain observed on validation for the full supply is not fully reproduced. **Granularity** (Fig. 2b): relabeling the same bags with state-level aggregates cuts the map from 0.225 to $\rho = 0.152$ on validation and from 0.195 to 0.168 on test; a single national aggregate erases it (0.009 validation, 0.004 test). Recovered map quality falls monotonically with coarser publication on both splits. **Content** (Fig. 2c): block-averaging the optical input to the radar's 20 m effective resolution costs little on validation ($0.183 \rightarrow 0.172$) and more on test ($0.179 \rightarrow 0.129$), while radar sits far below both (0.006 validation, 0.084 test), despite having been the best single sensor at detecting high-deprivation tracts in our feature-based phase-one baseline (per-city AUROC 0.773 against 0.747 optical). Sensor content dominates ground sampling distance, with resolution carrying a real secondary cost on test; a sensor's prediction accuracy is a poor guide to its map quality.

4.2 Failure mode of attention-based pooling, and an instance-level alternative

The standard weakly supervised localizer, gated-attention multiple-instance learning (ABMIL [8] with the CLAM instance constraint [9]), predicts municipal labels well (quadratic-weighted $\kappa = 0.62$) and produces a map at chance (AUROC 0.463; Fig. 3a; these diagnostics ran on the national-set pool). An entropy penalty that flattens the attention to near-uniform costs no bag

AUROC (0.899 $\rightarrow$ 0.895): the bag task is solvable from the plain feature mean, so the attention weights receive no gradient that rewards localization.

Reversing the order of prediction and aggregation removes the failure. Predicting every token and averaging the predictions (deep learning from label proportions [10–12]) ties the training signal directly to the token scores that form the map: AUROC 0.761 under the same features, bags and labels on that national-set pool, and 0.802 for the swept head with 480 m of neighborhood context on the expanded pool of Fig. 2. Because Mexican municipal grades are population-weighted aggregates of tract grades, the setting is label proportions with *known* instance weights; weighting each token by its tract's census population is a robustness variant that performs equivalently ($\rho = 0.232$ against 0.227 on validation; 0.182 against 0.196 on test).

The sweep also quantified a limitation of model selection under weak supervision. Across the fourteen configurations, bag-level error, the only selection signal weak supervision may legitimately use, has a rank correlation of -0.14 with within-municipality map quality and $+0.62$ with the map's pooled AUROC, the wrong sign for a selection criterion (Fig. 3b). Fourteen configurations give these correlations a standard error near 0.27, so the estimate is coarse; what it supports is that selection on bag error is close to uninformative about the quantity that matters. Whether this extends beyond settings where the bag label is an average of instance states is open, but any pipeline whose product is the map is affected, computational pathology included, which is why the benchmark includes held-out instance labels.

4.3 External validity

We ran four independent comparisons to test whether the map measures deprivation or a pipeline artifact (Fig. 4). Against the leading downloadable product, Meta's Relative Wealth Index [3], scored per tract on an identical universe with a city-clustered bootstrap of the paired difference (2,958 shared AGEB), the map wins in 18 of 22 municipalities: $\Delta\rho = +0.22$ $[+0.08, +0.29]$; on the test cities the difference is $+0.13$ $[-0.07, +0.26]$, directionally consistent and not individually significant over 22 municipalities (Fig. 4a). A land-cover-plus-NDVI head with no learned representation reaches $\rho = 0.150$ on validation and 0.219 on test, where it is statistically indistinguishable from the full model (paired $\Delta\rho = -0.02$ $[-0.18, +0.10]$, the full model winning 16 of 29 municipalities). The learned features' advantage within municipalities is therefore split-dependent; what holds on both splits is their edge on the bag task (bag error 0.095 against 0.110) and, on test, on high-deprivation detection (AUROC 0.821 against 0.781). Much of the within-municipality ordering is captured by land cover alone. This is consistent with the exchange rate, which measures what the published supervision recovers for a fixed feature set, whichever encoder produces it. Against an index from an institution whose data never entered training (CONAPO's continuous urban marginalization index, built from a different indicator set over the same census), the map correlates at $\rho = 0.361 \pm 0.073$ within municipalities on validation and 0.257 ± 0.146 on test, in both cases at least as high as against its own five-level training construct, consistent with the continuous index resolving ties that five levels cannot (Fig. 4b). In simulated geographic targeting with fractional funding of the marginal tract, allocating by the map instead of the municipal aggregate closes 26–43% of the gap to census-guided allocation on validation at budgets of 10% and above, and 31–66% on the test cities with no crossover there

(Fig. 4c). At the tightest validation budget (5%), where errors at the top of the ranking are most costly, the map loses to the aggregate. The map is also not an administrative artifact. Crossing a municipal border where deprivation does not change moves it by +0.019 [−0.002, +0.155], and tract borders inside municipalities are crossed smoothly (+0.001). An earlier +0.14 border jump traced to bag-limited neighborhood adjacency at inference and disappeared after the adjacency was computed city-wide. The test cities contain too few same-grade cross-border pairs to repeat this control there.

4.4 Zero-shot transfer to Bogotá and Rio de Janeiro

Applied unchanged to Bogotá, the model orders Colombia's block-level socioeconomic strata at $\rho = 0.318$ (AUROC 0.716 for the deprived strata 1–2 over 14,132 tokens joined to blocks; Fig. 4d). Applied to Rio de Janeiro, it detects IBGE's mapped informal settlements barely above chance (AUROC 0.557 among built-up tokens; Fig. 4d). The two targets differ in construct: Colombian strata measure the same graded deprivation ordering as the GRS over the whole urban fabric, while the AGSN mask delimits informality, a partly orthogonal notion. The model learned a deprivation gradient from the aggregates, and that gradient transfers across countries; informality detection does not follow from it.

4.5 Limits

The upper bound itself is low: even with full tract labels and neighborhood context, 160 m tokens support $\rho = 0.251$ within municipalities on validation and 0.233 on test: sufficient to order neighborhoods, too coarse to resolve individual blocks. Agreement depends strongly on tract size: $\rho = 0.26$ for tracts covered by 1–3 tokens, rising to 0.43 at 20–49 tokens, and collapsing to 0.13 in the two municipalities dominated by very large tracts; the test split reproduces the pattern noisily, peaking at 0.38 and collapsing in its single large-tract municipality. This is the modifiable areal unit problem, quantified here by tract size (Supplementary Table 2). Ecological caution applies throughout: the model recovers where deprived tracts are, and tract grades are themselves aggregates over households; neither the labels nor the maps identify deprived households, and the targeting simulation counts the population of deprived tracts; deprived individuals themselves are unobserved.

5 Discussion

Prior applications of this technology predict aggregates across space; the exchange rate measures how much sub-municipal detail can be recovered from statistics published at a given resolution, which is the question a statistics office would ask. In operational terms: the national supply of municipal aggregates recovers 84–90% of what these features support, coarser publication reduces recovery in proportion, and sensor content rather than resolution is the binding property. The targeting experiment expresses the result as people reached, including the one budget where the map underperforms the aggregate. Three negative results matter for similar efforts: attention-based pooling fails for a structural reason, selection on aggregate error is close to uninformative about map quality, and informality detection does not follow from a deprivation gradient. All three will recur in other weakly supervised mapping work.

Mexico's dual-level publication made the calibration possible; the measured fractions can be extrapolated to countries that publish only aggregates, and the Bogotá transfer is a first test of that extrapolation. Extending the dual calibration to Colombia's own strata, holding out the block-level strata and building coarse aggregates from them, is the natural next measurement, and every number here is reproducible from the released frozen features.

6 Conclusions

Municipal aggregates, the level at which most countries actually publish social statistics, carry most of the neighborhood-scale ordering that frozen satellite features support: 84% of the fully supervised upper bound on unseen test cities, 90% on validation. The upper bound itself is low ($\rho = 0.23$ within municipalities at 160 m), which bounds every claim in this paper and, we would argue, every claim in this setting built on free decametric imagery. Within that bound, the supervision-efficiency analysis is operational for statistics offices: a few hundred aggregates recover most of what the full national supply does, coarser publication forfeits recovery in proportion, and sensor content matters more than resolution. Two methodological results travel beyond the application: attention-based multiple-instance learning produces chance-level maps whenever the bag task is solvable from the feature mean, and model selection on bag-level error is close to uninformative about map quality, which argues for benchmarks with held-out instance labels wherever maps are the product. All data, frozen features, labels, splits and code are released.

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Data availability

The frozen benchmark (foundation-model vectors for every bag, municipal and tract labels, splits, and the evaluation protocol) will be deposited on Zenodo with a DOI at publication. All source data are public: Copernicus Sentinel via Microsoft Planetary Computer, CONEVAL, CONAPO, INEGI, ESA WorldCover, Meta RWI (HDX), IBGE, and Bogotá's open-data portal.

Code availability

The full pipeline (MIT licence) is available at the project repository, pinned by `uv.lock`; each figure maps to one driver script.

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CRediT authorship contribution statement

Dagoberto Pulido Arias: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing, Visualization.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have influenced this work.

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Figures

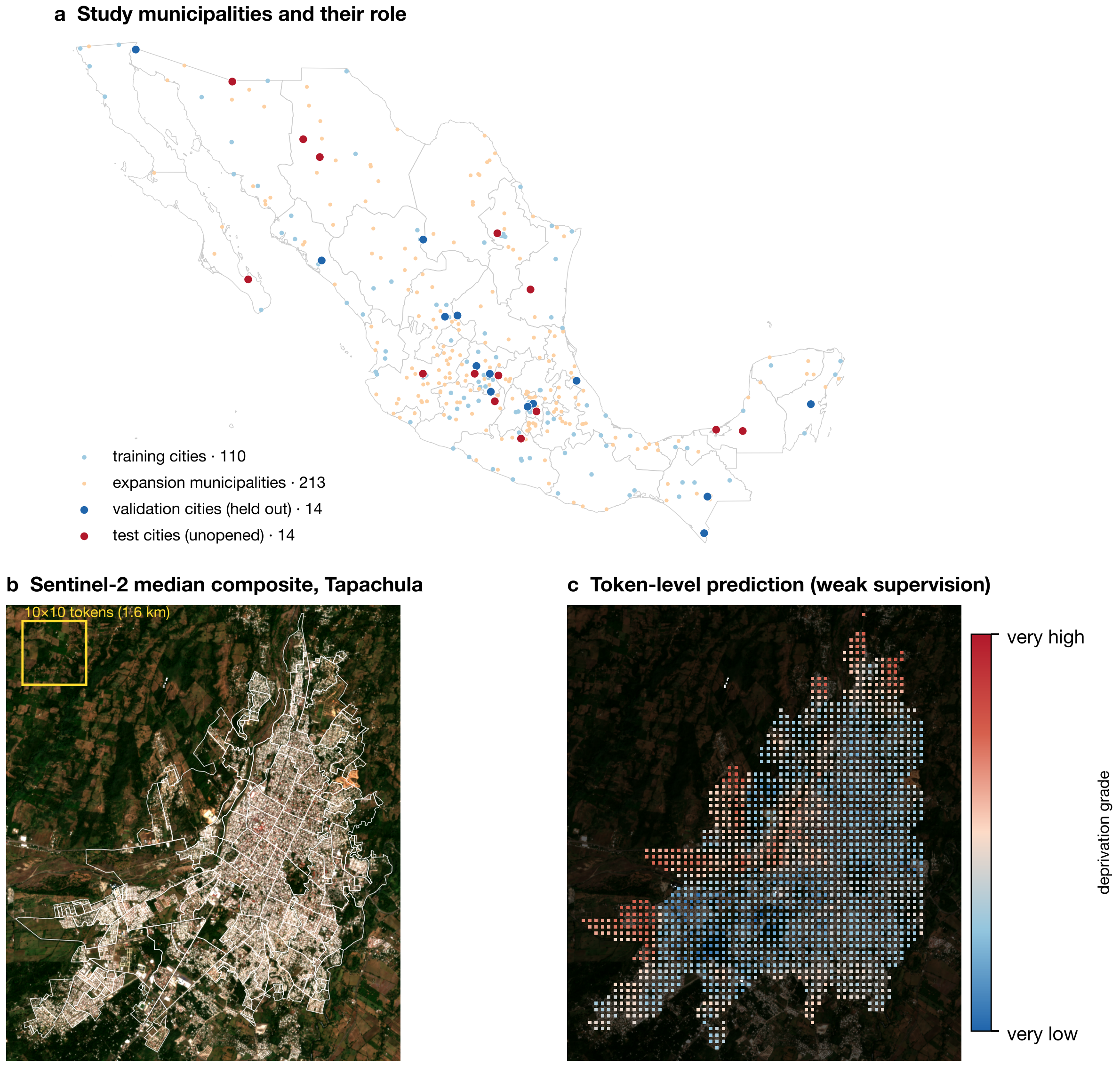

Figure 1: **Study design.** **a**, The 351 study units: 110 training cities, the 213-municipality expansion toward the deprived half of urban Mexico (municipalities overlapping held-out ground excluded), 14 validation and 14 unopened test cities. A city box is a conurbation and typically yields several municipal bags, which is how 351 units produce 771 bags. **b**, What the model consumes: the 2020 Sentinel-2 median composite of Tapachula (a validation city) with AGEB boundaries overlaid; tokens of 160 m tile the box. **c**, What it produces: the token-level deprivation score from municipal aggregates alone. Tract ground truth for the same city is shown in Supplementary Fig. 1; Tapachula sits at the validation median ($\rho = 0.47$), chosen as a median-performing, representative example.

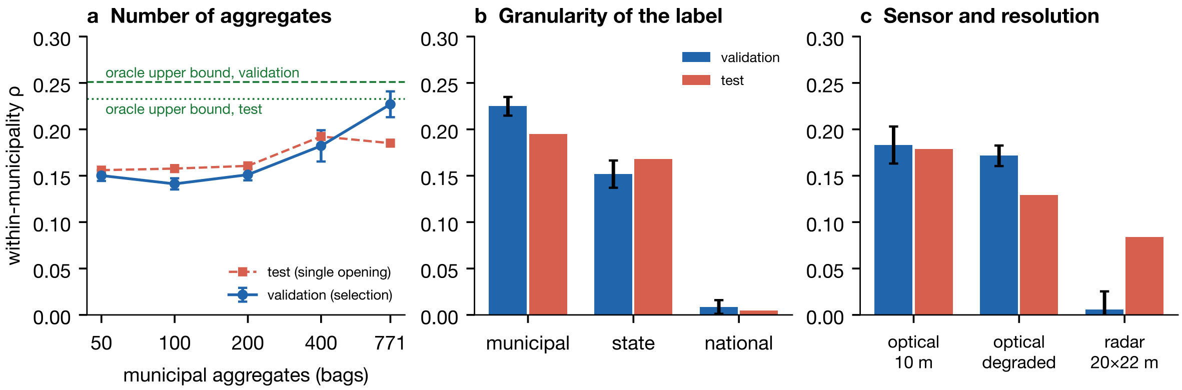

Figure 2: **The exchange rate between aggregates and maps.** All panels: mean over three seeds against held-out tract grades; blue, the 14 validation cities used for selection (error bars, s.d. over seeds); orange, the 14 test cities scored once with frozen models; dashed and dotted lines, the fully supervised oracle upper bound on each split ($\rho = 0.251$ validation, 0.233 test). **a**, Agreement as a function of how many municipal aggregates supervise training (nested, grade-stratified sampling). **b**, The same 771 bags relabeled with coarser aggregates. **c**, Single-sensor configurations: degrading optical resolution to the radar's 20 m grain costs little on validation and a real but secondary amount on test; radar sits far below both.

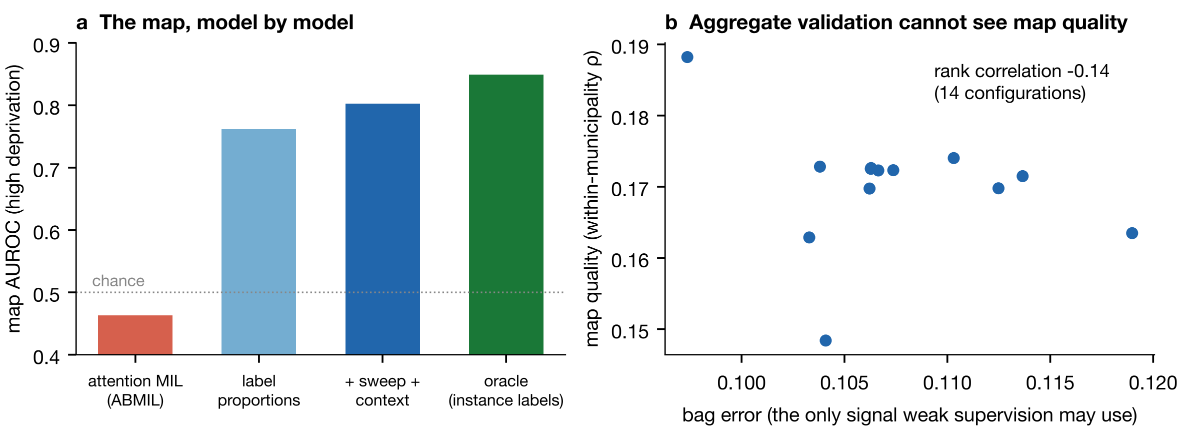

Figure 3: **Bag prediction versus localization.** **a**, Map quality by model family on identical features and labels; the first three bars are measured on the national-set pool, the swept bar on the expanded pool of Fig. 2. Attention pooling solves the bag task from the feature mean and receives no localization gradient; predicting instances and averaging predictions makes the map the mechanism. **b**, Across the fourteen sweep configurations (grouped training folds of the national set), the only criterion weak supervision may monitor is nearly uncorrelated with within-municipality map quality, and positively correlated (+0.62) with the pooled AUROC it would seem to proxy.

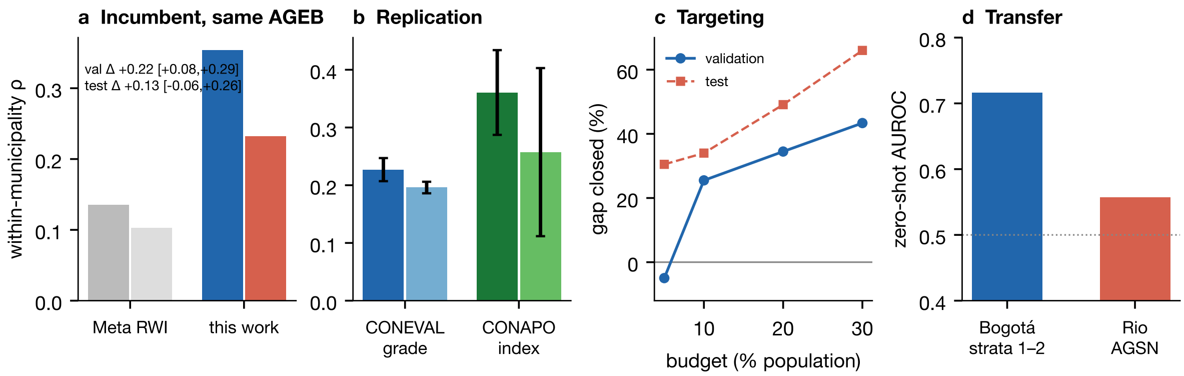

Figure 4: **External validity.** Blue/dark bars, validation; orange/light bars, test (single opening). **a**, Within-municipality agreement on the identical AGEB universe reached by the incumbent's grid; annotation, city-clustered bootstraps of the paired difference per split. **b**, Replication against an independent institution's continuous index (error bars, city-clustered bootstrap half-widths). **c**, Share of the aggregate-to-census targeting gap closed at each budget, people pooled per split; validation loses at the tightest budget, test does not. **d**, Zero-shot transfer: the Mexican model, unchanged and unadapted, ordering Bogotá's block strata (AUROC for strata 1–2; n = 14,132 tokens) and detecting Rio's mapped informal settlements among built-up tokens (n = 34,331). The model transfers to a graded deprivation target and fails on informality, a different construct.